Technical Report: FastSpeech 2 TTS Implementation (English & Telugu)

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1. Implementation Approach: Implemented FastSpeech 2 architecture with Feed-Forward Transformer blocks and custom Variance Adaptor for Duration, Pitch, and Energy predictions. G2P was utilized for English (Svarah) and Indic-NLP syllabic alignment for Telugu (IndicTTS).

2. Training & Convergence: Both models demonstrated stable convergence across 50 production epochs using AdamW and Cosine Annealing learning rate schedule.

3. Evaluation: Evaluated thoroughly on benchmark sentences (E1-E4 and T1-T4) demonstrating natural prosody, phonetic durations, and accurate intonation contours.